CS1200: Intro. to Algorithms and their Limitations Profs. Anshu and Nesson

Problem Set 0

Harvard SEAS - Fall 2026 *Due: Sat 2026-09-08 (11:59PM)*

Please see the syllabus for the full collaboration and generative AI policy, as well as information on grading, late days, and revisions.

All sources of ideas, including (but not restricted to) any collaborators, AI tools, people outside of the course, websites, ARC tutors, and textbooks other than Hesterberg–Vadhan must be listed on your submitted homework along with a brief description of how they influenced your work. You need not cite core course resources, which are lectures, the Hesterberg–Vadhan textbook, sections, SREs, problem sets and solutions sets from earlier in the semester. If you use any concepts, terminology, or problem-solving approaches not covered in the course material by that point in the semester, you must describe the source of that idea. If you credit an AI tool for a particular idea, then you should also provide a primary source that corroborates it. Github Copilot and similar tools should be turned off when working on programming assignments.

If you did not have any collaborators or external resources, please write 'none.' Please remember to select pages when you submit on Gradescope. A problem set on the border between two letter grades cannot be rounded up if pages are not selected OR collaborators not noted.

Your name: Chiamaka (Amaka) Ihejirika
Collaborators and External Resources: Manasa Bala
No. of late days used on previous psets: 1
No. of late days used after including this pset: 7

The purpose of this problem set is to reactivate your skills in proofs and programming from CS20 and CS32/CS50. For those of you who haven't taken one or both those courses, the problem set can also help you assess whether you have acquired sufficient skills to enter CS1200 in other ways and can fill in any missing gaps through self-study. Even for students with all of the recommended background, this problem set may still require a significant amount of thought and effort, so do not be discouraged if that is the case and do take advantage of the staff support in section and office hours.

For those of you who are wondering whether you should wait and take CS20 before taking CS1200, we encourage you to also complete the CS20 Placement Self-Assessment. Some problems there that are of particular relevance to CS1200 are Problems 2 (counting), 4 & 5 (comparing growth rates), 9 (quantificational logic), and 12 (graph theory).

Written answers must be submitted in pdf format on Gradescope. Although LaTeX is not required, it is strongly encouraged. You may handwrite solutions so long as they are fully legible. The `ps0` directory, which contains your code for problems 1a and 1c, must be submitted separately to an autograder on Gradescope.

To obtain the starter code for the programming problems, please download it from GitHub.

1. (Binary Trees)

   In the cs1200 GitHub repository, we have given you a Python implementation of a binary tree data structure, as well as a collection of test trees built using this data structure. We

specify a binary tree by giving a pointer to its *root*, which is a special *vertex* (a.k.a. *node*), and giving every vertex pointers to its *children* vertices and its *parent* vertex as well as an identifying *key*:

```
class BinaryTree:
    def __init__(self, root):
        self.root: BTvertex = root

class BTvertex:
    def __init__(self, key):
        self.parent: BTvertex = None
        self.left: BTvertex = None
        self.right: BTvertex = None
        self.key: int = key
        self.size: int = None
```

to id the node

no object as parent or child yet

In CS50, the concept of a Python `class` was not covered. Here, with `BinaryTree` and `BTvertex`, we are using them in the same way as a `struct` in C. An object `v` of the `BTvertex` class contains five attributes, which we list with the type of the object we expect to be named by each attribute (using the Python type annotation syntax). These attributes can be accessed as `v.parent`, `v.left`, `v.right`, `v.key`, and `v.size`. For example, `v.left.key` is the key associated with `v`'s left child. An object of the `BinaryTree` class contains only one attribute, which is the `BTvertex` object that is the root of our binary tree. You can create a `BinaryTree` object as follows:

key

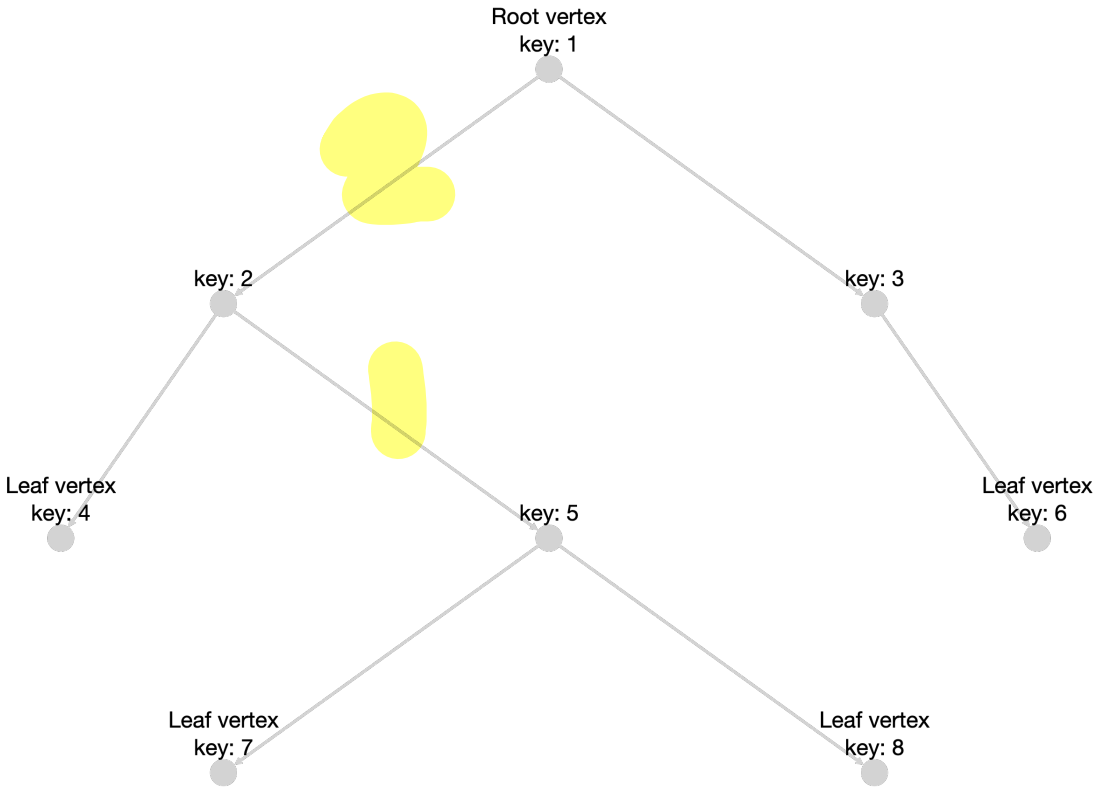
```
root = BTvertex(1200)
tree = BinaryTree(root)
tree.root.left = BTvertex(1210)
tree.root.right = BTvertex(1240)
```

You can then print attributes of the newly created `BinaryTree` object:

```
print(tree.root.key)
>> 1200
print(tree.root.left.key)
>> 1210
```

Classes are more general than structs because they can also have private attributes and methods that operate on the attributes, allowing for object-oriented programming. However, you won't need that generality in this problem set.

Here is an instance `T` of `BinaryTree`:



defining a Binary Tree again

A `BinaryTree` `T` contains only a pointer to its root vertex, `T.root`, which is required to satisfy `T.root.parent==None`. In the above example, the root is the vertex with key 1 (i.e. `T.root.key==1`). A binary tree vertex `v` can have zero, one, or two children, determined by which of `v.left` and `v.right` are equal to `None`. In the above example, the vertex `v` with key 3 has `v.left==None` but `v.right` is the vertex with key 6. A *leaf* is a vertex with zero children, i.e. `v.left==v.right==None`.

A vertex `w` is *descendant* of a vertex `v` if there is a sequence of vertices $\mathtt{v}_0, \mathtt{v}_1, \ldots, \mathtt{v}_k$, $k \in \mathbb{N}$ such that $\mathtt{v}_0 = \mathtt{v}$, $\mathtt{v}_k = \mathtt{w}$, and $\mathtt{v}_i \in \{\mathtt{v}_{i-1}\texttt{.left}, \mathtt{v}_{i-1}\texttt{.right}\}$ for $i = 1, \ldots, k$.[1] In the above example, the vertex with key 5 is a descendant of the root (with a path of length 2), but is not a descendant of the vertex with key 3. The sequence $\mathtt{v}_0, \mathtt{v}_1, \ldots, \mathtt{v}_k$ is called a *path* from `v` to `w` and k is the *distance* from `v` to `w`. Taking $k = 0$, we see that `v` is a descendant of itself.

The *vertex set* of a binary tree `T` consists of all of the descendants of `T.root`. The *size* of `T` is its number of vertices. The *height* of `T` is the largest distance from the root to a leaf. The above example has size 8 and height 3.

Given any vertex `v` in a tree, the *subtree* rooted at `v` consists of all of `v`'s descendants. Note that we can remove a subtree and turn it into a new tree `S` by setting `S.root=v` and `v.parent=None`.

For now, the `key` attribute serves to distinguish vertices from each other in our tests and help illustrate what the algorithms are doing. The `BTvertex` class also has a `size` attribute,

[1] $\mathbb{N}$ denotes the natural numbers $\{0, 1, 2, 3, \ldots\}$. Since we are computer scientists, we start counting at 0.

which is initialized to `None` in all of the test instances; it will be filled in by the program you write in Part 1a.

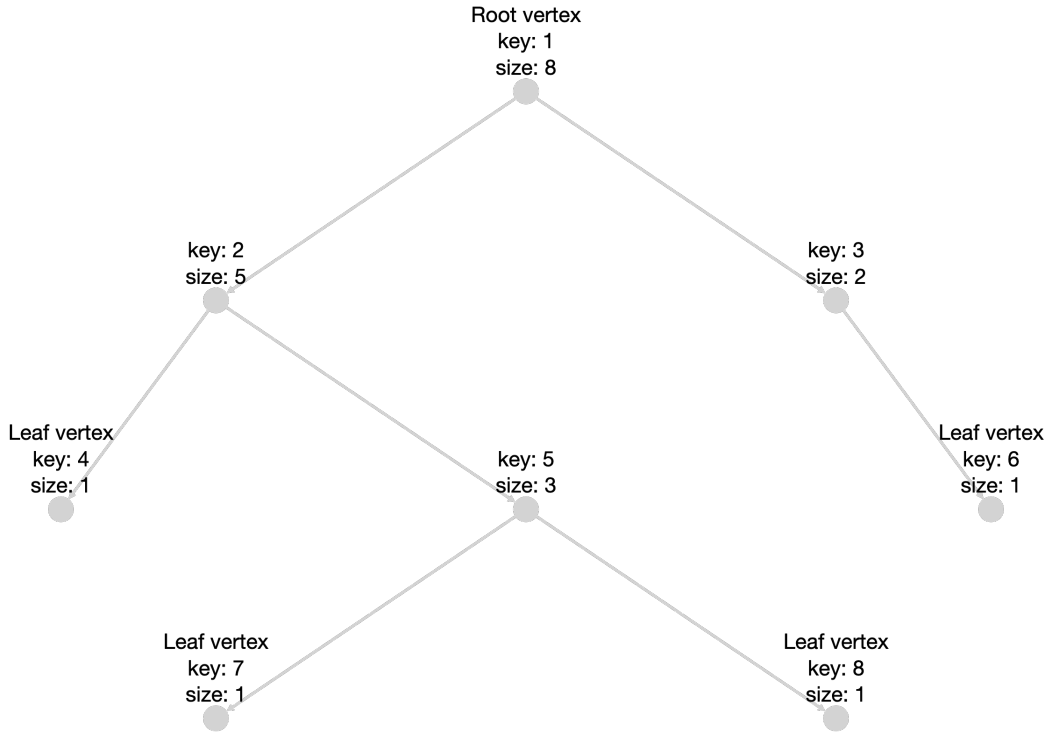
An instance `T BinaryTree` is *valid* if it satisfies the following constraints:

- `T.root.parent==None`
- `T` has finitely many vertices.
- No two vertices `v`, `w` of `T` share a child, i.e. $\{\texttt{v.left}, \texttt{v.right}\} \cap \{\texttt{w.left}, \texttt{w.right}\} = \emptyset$.

All of the test instances we provide are valid, and furthermore have the property that all of the vertices have distinct keys (which is something we often want, but not always).

(a) (recursive programming) Write a recursive program `calculate_sizes` that given a vertex `v` of a binary tree `T`, calculates the sizes of all of the subtrees rooted at descendants of `v`. After running your program on `T.root`, every vertex `v` in `T` should have `v.size` set to the size of the subtree rooted at `v`. (Recall that the size attributes are initialized to `None`.) We call the resulting tree a *size-augmented* tree.

For example, if `T` is the tree shown above, then calling `calculate_sizes(T.root)` should modify `T` to be the following size-augmented tree:



Your program should run in time $O(n)$ when given the root of a tree with n vertices. In a sentence or two, informally justify why your program has such a runtime. on next page

(b) (proofs by induction) Let t be a positive integer that is fixed for this question. Prove that every binary tree `T` of size at least $2t+1$ has a vertex `v` such that the subtree rooted at `v` has size at least t and at most $2t-1$. (Hint: use strong induction on the size n of

Your program should run in time $O(n)$ when given the root of a tree with n vertices. In a sentence or two, informally justify why your program has such a runtime.

```python
def calculate_sizes(v):
    '''
    include urself if u exist as well as the number of children
    that your children have.
    recursive.
    '''
    # base case
    if v is None:
        return 0
    # size includes self and child sizes
    v.size = 1 + calculate_sizes(v.left) + calculate_sizes(v.right)
    return v.size
```

rec steps

Each node is visited exactly once since no two nodes can share a child and node exploration goes from parents to child. Because there are n ~~exit~~ nodes and each operation done (addition & field access) has time O(1), the total program has a runtime of O(n).

(b) (proofs by induction) Let t be a positive integer that is fixed for this question. Prove that every binary tree T of size at least $2t+1$ has a vertex v such that the subtree rooted at v has size at least t and at most $2t-1$. (Hint: use strong induction on the size n of T, starting with base case $n = 2t+1$. Remember that nodes in binary trees can have 2, 1, or 0 children.)

Rewording: Prove that a vertex v exists in a tree T of size 2t + 1 such that t <= v.size <= 2t - 1 << call this P(n).
Base Case:
Let n = 2t + 1, meaning the root, a, of T has size 2t + 1. Let b and c be the children of a (and thus subtrees of T). We know that because a accounts for the 1 in 2t + 1, b.size + c.size must equal 2t. This means that at least b.size or c.size have to be at least t because if neither of them are, then b.size + c.size < 2t (since t - 1 + t - 1 = 2t - 2 which contradicts b.size + c.size = 2t). This means that the lower limit of the proof (a subtree of at least size t exists) is met.

As for the upper limit, if either b.size and c.size are greater than 2t - 1, assuming that both children exist (b.size and c.size >= 1), then b.size + c.size > 2t, which contradicts b.size + c.size = 2t. However, if only child b exists, then b.size CAN be 2t, which would automatically mean that b's children (b.left and b.right) would have a combined size of 2t - 1, thus satisfying the claim that there exists at least one subtree of T with a size no greater than 2t-1.

Induction Hypothesis: Assume that P(m) is true for every tree with 2t + 1 <= m <= k number of nodes.

Induction Step: Let T be a random binary tree with k+1 nodes, we set out to prove P(k+1). Let's call the root A, wherein A.size = k+1.
- We know that the sum of the sizes of A's children will be k (B.size + C.size = k), and we know that k >= 2t + 1. Therefore, at least one of the children, B and C, has a size of at least t by the inductive hypothesis.

- Looking at this child, let's say it's child B, B.size can be
(1) exactly inside of our desired range of less than 2t - 1 but greater than t, meaning P(k+1) holds here.
(2) If B.size is 2t (outside the range), then we can simply look to one of it's children which we already know one of them must be greater than t (since the total child size sum must be 2t - 1 and if both are less than t then the sum would be 2t - 2) and we know that the child size sum is 2t - 1 meaning neither of them can be greater than 2t - 1, so P(k+1) holds here too.
(3) Furthermore, if B.size >= 2t + 1, since we know that B.size <= k and that k >= 2t + 1, then by the induction hypothesis, we know that B contains a subtree with a size between t and 2t - 1.
Thus, P(k+1) holds.

Conclusion:
Therefore, we proved by strong induction that any binary tree with 2t + 1 nodes, where t is a fixed positive int, contains a subtree rooted at v wherein the size of that tree is at least t and at most 2t - 1.

(c) (from proofs to algorithms) Turn your proof from Part 1b into a Python program `FindDescendantOfSize(t,v)` that, given a positive integer `t` and a vertex `v` of a *size-augmented* tree `T` such that `v.size` $\geq$ `2t+1`, finds and returns a descendant `w` of `v` such that `t` $\leq$ `w.size` $\leq$ `2t-1`. Your algorithm should run in time $O(h)$, where h is the height of the subtree rooted at `v`; explain in words why this is the case.

```python
def FindDescendantOfSize(t, v):
    '''
    t is a POS INT, v is a "SIZE AUGMENTED" TREE st v.size >= 2t + 1.
    Find subtree with root w wherein t <= w.size <= 2t - 1.
    Should use calculate_sizes methinks. OOHHHH already size augmented
    '''
    # removed base cases (v is none and v.size is in range because they never hit given the
    # inputs since is always at least 2t + 1)

    # Check child size is over or perfectly already in range because
        # sum of child sizes is at least 2t so one child must be at least t
    if v.left is not None:
        # just out of range
        if (v.left.size >= 2*t):
            return FindDescendantOfSize(t, v.left)
        # if less than t ignore and go to right child
        elif (v.left.size >= t):
            return v.left

    if v.right is not None:
        # just out of range
        if (v.right.size >= 2*t):
            return FindDescendantOfSize(t, v.right)
        elif (v.right.size >= t):
            return v.right
```

Each recursive call from the first root v is to one of its children. This means that each call is walking the path down from the root until it gets to what we know to be the desired child because the tree is size augmented and thus the program stops immediately at the child h levels away whose size field fits our range. Meaning that we don't have to traverse the entire tree and all its levels in order to get to the root w. Furthermore, each of the comparisons are done in O(1) time.

Therefore, the total algorithm runtime is O(h) where h is the height of the subtree rooted at v and ending at w.

answered above

`T`, starting with base case $n = 2t + 1$. Remember that nodes in binary trees can have 2, 1, or 0 children.)

(c) (from proofs to algorithms) Turn your proof from Part 1b into a Python program `FindDescendantOfSize(t,v)` that, given a positive integer `t` and a vertex `v` of a *size-augmented* tree `T` such that `v.size` $\geq$ `2t+1`, finds and returns a descendant `w` of `v` such that `t` $\leq$ `w.size` $\leq$ `2t-1`. Your algorithm should run in time $O(h)$, where h is the height of the subtree rooted at `v`; explain in words why this is the case.

2. (asymptotic notation) Recall the definitions of asymptotic notation from CS20: Let f and g be functions from $\mathbb{N}$ to $\mathbb{R}^+$, where $\mathbb{N} = \{0, 1, 2, \ldots\}$ denotes the natural numbers (including 0, since we are computer scientists) and $\mathbb{R}^+$ denotes the positive real numbers.

   - We say that $f = O(g)$ if there is a constant $c > 0$ such that $f(n) \leq c \cdot g(n)$ for all sufficiently large n.
   - We say that $f = o(g)$ if for *every* constant $c > 0$, $f(n) \leq c \cdot g(n)$ for all sufficiently large n; equivalently, $\lim_{n\to\infty}(f(n)/g(n)) = 0$.

   (a) For each of the following pairs of functions, determine whether $f = O(g)$ and whether $f = o(g)$. Justify your answers.

   i. $f(n) = 3\log_2^3 n$, $g(n) = n^2 + 1$.
   ii. $f(n) = |\{S \subseteq [n] : |S| \leq 5\}|$, where $[n] = \{0, 1, 2, \ldots, n-1\}$, and $g(n) = 3n^3$.
   iii. $f(n) = 5^n$, $g(n) = n!$.
   iv. Prove or disprove: For all functions $f, g : \mathbb{N} \to \mathbb{R}^+$, $f = O(g) \Rightarrow \big(g = o(f)$ OR $f = o(g)\big)$.

3. (reflection: excitement and apprehension) On every problem set in cs1200, there will be a qualitative reflection question, usually about some aspect of your engagement with the course (but sometimes about the course content itself). Your answers to these questions, the in-class sender-receiver exercises and section attendance will be the basis of your participation grade. For starters, every thoughtful answer with some specifics will receive an R grade; R+ will be reserved for exceptional ones and grades below R for responses that are incomplete or generic.

On last page

Quick note on grading: Good responses are usually about a paragraph, with something like 7 or 8 sentences. Most importantly, please make sure your answer is specific to this class and your experiences in it. If your answer could have been edited lightly to apply to another class at Harvard, points will be taken off.

Note: As with the previous psets, you may include your answer in your PDF submission, but the answer should ultimately go into a separate Gradescope submission form.

(Re)read the syllabus. What aspect of cs1200 are you most excited about and what aspect are you most apprehensive about? Please address both excitement and apprehension in your answer, but it is also fine to say "I'm not excited about anything" or "I'm not apprehensive about anything." What efforts can you make as a student to make the most out of your excitement, and to help address your apprehension?

4. Once you're done with this problem set, please fill out this survey so that we can gather students' thoughts on the problem set, and the class in general. It's not required, but we really appreciate all responses!

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(a) For each of the following pairs of functions, determine whether $f = O(g)$ and whether $f = o(g)$. Justify your answers.

i. $f(n) = 3\log_2^3 n$, $g(n) = n^2 + 1$.

ii. $f(n) = |\{S \subseteq [n] : |S| \leq 5\}|$, where $[n] = \{0, 1, 2, \ldots, n-1\}$, and $g(n) = 3n^3$.

iii. $f(n) = 5^n$, $g(n) = n!$.

iv. Prove or disprove: For all functions $f, g : \mathbb{N} \to \mathbb{R}^+$, $f = O(g) \Rightarrow \big(g = o(f)$ OR $f = o(g)\big)$.

i) $\frac{3\log_2^3 n}{n^2+1}$ $f(n)$ / $g(n)$ — n^2+1 redundant. As $n\to\infty$, $f(n) \ll g(n)$, $\forall \Rightarrow \exists$ $\frac{\text{small}}{\text{BIG}} = 0$, meaning $f = o(g)$ AND $f = O(g)$

ii) f is a set S $\{0,1,\ldots,n-1\}$ where the abs. size is ≤ 5.

g is $= 3n^3$; influential

$f(n) =$ is $\binom{n}{0} + \ldots + \binom{n}{5}$ bc ≤ 5

$\binom{n}{5} = \frac{n(n-1)(n-2)(n-3)(n-4)}{120}$ most influential $\approx \frac{n^5}{120} = f(n)$

$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{\frac{1}{120}n^5}{3n^3} = n^2 \to \infty$

so $f \neq o(g)$ & $f \neq O(g)$

iii) $n \geq 0$ and $5^n \ll n!$ so $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{\text{small}}{\text{BIG}} = 0$

as $n\to\infty$ so, $f = o(g)$ AND $f = O(g)$

iv) False Prove w/ counterclaim that $f = O(g)$, $f \neq o(g)$, $g \neq o(f)$

Set $f(n) = g(n) = n$

$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{n}{n} = 1$ <- Const num. so f is never strictly growing slower than g Meaning $f = O(g)$ but not $o(g)$

If we flip the fraction we get $\lim_{n\to\infty} \frac{g(n)}{f(n)} = \frac{n}{n} = 1$ Proving that $g \neq o(f)$ either ↓

Therefore, since there exists a function st $f = O(g)$ but not $f = o(g)$ OR $g = o(f)$, we prove the claim to be ~~FALSE~~

Note: As with the previous psets, you may include your answer in your PDF submission, but the answer should ultimately go into a separate Gradescope submission form.

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Took a look at the syllabus and it doesn't really say what exactly we'll be learning week to week, so I can't say that I'm particularly excited about anything. However, I am a bit apprehensive about my time management (considering that I'm already taking a late day for ps0) and how I'll manage to find time to go to OH when I run into problems with homework. Ultimately though, I'm not too worried since there are many TFs and thus many office hours for me to attend. Since there's nothing off the linked syllabus for me to get excited about, I figure that my excitement for the class will have to come from our main lectures. As a student, I think that the way to maximize my excitement would be to try and follow along as best as I can in class (some lectures tend to be fast paced and I struggle to fully grasp a concept the slides move along). Furthermore, since I'm apprehensive about lecture pace, I can try to find sections that fit into my schedule so that I can practice/learn/reinforce concepts that were gone over in class. If things get to bad, I can always reach out to the Patel Fellow or even to course staff if the fellow's calendar is booked.